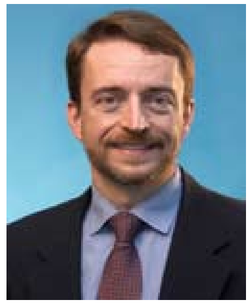
Intel's senior vice president Pat Gelsinger would like to herald a new era: "Our multi-decade old 3D graphics rendering architecture that's based on a rasterization approach is no longer scalable and suitable for the demands of the future".

That 'fixed-function unit', nestled within a Larrabee processor will eventually take the form of an execution unit dedicated to the task of raytracing. This small detail is the source of the brewing war between Intel and graphic vendors; as well as a point of disbelief within the game developer community.

Raytracing is not just another way to do things. It is a completely different way to do things. It is a 180-degree turn to an industry that has long accepted rasterisation as the approach that makes the most sense.



Pat Gelsinger: "Graphics that we have all come to know and love today...It's coming to an end"

So What Is Raytracing?

Imagine this scene before you: you are holding and reading this magazine. Now imagine your eye as the "camera" in a 3D game—the camera is the object that is doing the viewing, while the magazine is the scene being viewed. In raytracing an "eye ray" or a "primary ray" is projected from the viewpoint of the camera. Ray tracing uses physics simulations of propagation of rays to render an object. The algorithm first shoots the primary ray from the perspective of the eye. It then determines which object is hit first on the path of the ray. In this example, that object would be the magazine. At this point the magazine's material will determine the behaviour of the primary ray. If the material is transparent, the ray will pass through. If it is a mirrored surface, the ray will be reflected (angle of reflection, etc. calculated by the raytracing algorithm). If the material is a slightly glossy magazine paper—it will reflect some and absorb some, and so on. The raytracing algorithm also determines if the object it hits is in shadow. To do this, it shoots off a ray from the magazine towards the source of light—if this ray hits the source, then the magazine can "view" the light source and is therefore lit. If the ray is obstructed, the magazine is under a shadow.

This is of course, an oversimplification of the process. The take-away information is that raytracing takes a holistic look at the entire scene that is to be rendered, whereas a "rasterization" approach basically breaks down the entire scene into billions of component triangles. Due to the means used, the cost of raster graphics processing is linear with the number of pixels to be processed. On the raytracing side: the cost of raytracing increases linearly with the number of rays shot.

Intel's research into raytracing showed an interesting behaviour. If the scene being rendered is kept static, then there is a point at which software-assisted raytracing technique is almost as fast as a hardware accelerated raster technique. Software assisted implies CPU driven. Furthermore, the research found that if multiple processors are thrown at raytracing, the performance scales exponentially. Multiple processors became multi-core processors and will eventually become many-core processors.

John Carmack : "...while everybody thinks [raytracing] is going to be great, I have to reiterate that nobody has actually shown exactly how it's going to be great"

Intel thus envisions a raytracing technique to render graphics, instead of a raster approach. If you consider the architecture of the Larrabee processor—16 to 24 cores, each running at 1.7GHz at least, along with a dedicated unit to accelerate some raytracing calculations and helped by an extended vector instruction set—you can see why raytracing is the future for Intel.

A Muddy Path

While Intel would like the entire industry to embrace raytracing as the future, ground-realities are different. The largest hurdle is that game developers do not see anything wrong with the contemporary rasterization approach to graphics. Neither do the GPU vendors of today. The current ecosystem has embraced raster graphics and has evolved to accommodate its idiosyncrasies. The ray-trace only path proposed by Intel has few backers outside of the company.

Cevat Yerli the CEO of Crytek (*FarCry* and *Crysis*), doesn't think that there is a "compelling example for using pure classical ray tracing". He opines that "...there are a variety of graphics problems which would suit a hybrid solution of rasterization and raytracing, and [that] most likely is the way to go." His view is reflected by Intel itself—the initial builds of Larrabee will be derived from Intel's G45 integrated graphics. This indicates that the early Larrabee variants will not be for high-end graphics processing, but will be more of an integrated graphics replacement—perhaps a notebook part.

The next step would then be a slight raytracing nudge to produce eye-candy either not possible or prohibitively costly, using traditional rasterization methods: effects such as detailed reflections, refractions, and shadows, perhaps. Intel's goal could then be to slowly wean the industry on its diet of raytracing extensions, to eventually replace raster hardware accelerators by its own breed of processors.

John Carmack of id Software echoes this sentiment—"No matter who does what, the next generation is going to be really good at rasterization—that is a foregone conclusion. Intel is spending lots of effort to make sure Larrabee is a competitive rasterizer."

Carmack also brings up an interesting point of 'show, not tell', when he says that "while everybody thinks [raytracing] is going to be great, I have to reiterate that nobody has actually shown exactly how it's going to be great. The best way to evangelize your technology is to show somebody something... to kind of eat your own dog food, in terms of working with everything." Intel recently made two important purchases—the Havok physics engine, and Project Offset game engine. Perhaps these purchases are meant to do exactly what John Carmack proposes—to show the development community exactly why raytracing is the way ahead.

Meanwhile, Intel faces off against an entire industry. Who will do the actual ass-whooping; remains to be seen ■

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